

Literature review & Mechanisms definition

Comparing humanitarian event, report, rainfall and market data across the Horn of Africa

Samuel Aracena | MSc Data Science | Middlesex University
Study scope: Ethiopia, Somalia, South Sudan and Kenya · 2015–2024

Actions for this week

- Perform a broad literature review on humanitarian prediction (not only food insecurity).
- Identify which features are commonly used by humanitarian researchers.
- Define the final unit of analysis (e.g., one row = one ADM2 district × one month)
- Choose the core datasets (e.g., IPC, ACLED, WFP Food Prices, ReliefWeb or GDELT, etc)
- Define 5–6 humanitarian mechanisms and map each source to them.



Literature review

Paper Review: Pushing the Boundaries of Anticipatory Action Using Machine Learning (Kjærøum & Madsen, 2025)

Humanitarian Mechanism	Features
Conflict and Insecurity	Violent events, kidnappings, lootings, attacks by extremist groups, fatalities, violence against civilians
Food Security	Percentage of population in Cadre Harmonisé Phase 3 or above
Market Conditions	Food prices
Climate and Environmental Stress	Vegetation Health Index (VHI)
Development	Child mortality rate
Protection	Community protection indicators (e.g., perceived safety, freedom of movement, access to services)

- Kjærøum and Madsen (2025) developed the **AHEAD model**, a machine learning model that predicts conflict-related displacement several months in advance. The model combines data from multiple humanitarian sources to help humanitarian organisations prepare for future crises. It uses a **Bayesian state-space model** to estimate changes in displaced populations over time and compares its performance with a national forecasting model based on ensemble machine learning techniques.

Paper Review: Paper Review: Forecasting Trends in Food Security with Real-Time Data

Humanitarian Mechanism	Features Used
Historical Food Security	Previous prevalence of insufficient food consumption, previous reduced Coping Strategy Index (rCSI), Food Consumption Score (FCS)
Conflict and Insecurity	Number of conflict events, political violence indicators, fatalities (ACLED)
Climate and Environmental Conditions	Rainfall, Normalised Difference Vegetation Index (NDVI), drought and vegetation indicators
Market Conditions	Staple food prices, market prices, food price inflation
Population Movement	Displacement and migration indicators (where available)
Seasonality	Month of the year and seasonal patterns

- This paper presents a machine learning framework for predicting short-term food insecurity using real-time humanitarian and socio-economic data. The authors compare several forecasting methods, including **ARIMA**, **XGBoost**, **LSTM**, **CNN** and **Reservoir Computing**. They found that machine learning models perform better than traditional statistical methods, with Reservoir being the best one.

Paper Review: ViEWS: A Political Violence Early-Warning System (Hegre et al., 2019)

Humanitarian Mechanism	Features Used
Historical Conflict	Previous conflict events, previous fatalities, conflict persistence, recent violence
Spatial Diffusion	Conflict activity in neighbouring administrative areas, regional conflict intensity
Socio-economic factors	GDP per capita, demographic characteristics, education, long-term socioeconomic conditions
Political Context	Political institutions, elections, regime characteristics, political instability
Environmental Conditions	Rainfall variability, drought and selected climate indicators
Temporal Dynamics	Lagged conflict variables, previous conflict trends

- Hegre et al. (2019) developed the **Violence Early-Warning System (ViEWS)**, a machine learning framework that predicts the risk of organised political violence at national and subnational levels. The system combines several statistical and machine learning models into an **ensemble** to improve prediction accuracy. It generates monthly forecasts using information on historical conflict, political and socioeconomic conditions, environmental factors, and neighbouring regions. The model is designed to support early warning and humanitarian planning.

Paper Review: A Review on Machine Learning Models for Drought Monitoring and Forecasting (Osman et al., 2025)

Humanitarian Mechanism	Common Features	
Precipitation	Monthly precipitation, rainfall anomalies	Osman et al. (2025) provide a systematic review of machine learning methods used for drought monitoring and forecasting, rather than introducing a new prediction model. The review examines studies published between 2000 and 2025 covering meteorological, hydrological and agricultural drought. It highlights that many different machine learning models can successfully predict drought, but concludes that selecting the right input features is just as important as choosing the prediction model itself.
Temperature	Maximum, minimum and average temperature	
Atmospheric Conditions	Humidity, wind speed, evapotranspiration	
Hydrological Conditions	Streamflow, river discharge, groundwater levels	
Soil and Vegetation	Soil moisture, vegetation indices (NDVI), crop conditions	
Climate Indicators	SPI, SPEI, Palmer Drought Severity Index (PDSI), climate oscillation indices	

Unit of analysis

- For this study, the proposed unit of analysis is:

One row = one Admin 2 district × one month

This means that each row in the final dataset will describe the humanitarian conditions in one district during one calendar month.

Admin 2 was selected because it keeps more local detail than national or Admin 1 data. This is useful because neighbouring districts can experience very different levels of conflict, rainfall, food prices or humanitarian activity. Monthly aggregation was chosen because it provides enough temporal detail while still matching the frequency of most of the datasets.

Defining Mechanisms

Conflict and Insecurity

Category	Description
Source Dataset(s)	ACLED, GDELT and ICEWS
Raw Variables	Event date, location, event type and event category
Transformation Rule	Spatial join to Admin 2 districts, aggregate by month and calculate total event counts and counts by category for each dataset.
Engineered Features	Event count; Event counts by category
Expected Relevance	Measures conflict intensity affecting agriculture, livelihoods, markets and humanitarian access.
Known Limitations	Reporting bias, difference in news coverage, different event taxonomies and varying geolocation accuracy.

Climate and Agricultural Stress

Category	Description
Source Dataset(s)	CHIRPS
Raw Variables	Monthly rainfall (mm)
Transformation Rule	Calculate mean rainfall for each Admin 2 district using zonal statistics.
Engineered Features	Mean monthly rainfall
Expected Relevance	Represents environmental conditions influencing crop production and water availability.
Known Limitations	Rainfall alone does not capture complete agricultural conditions, as this might also be influenced by factors such as soil characteristics, irrigation practices and farming techniques.

Market and Food Price Pressure

Category	Description
Source Dataset(s)	WFP Food Prices
Raw Variables	Commodity, market, observation date, price and unit
Transformation Rule	Assign markets to Admin 2, aggregate monthly and calculate commodity price changes or a food basket index.
Engineered Features	Commodity monthly price changes OR Food Basket Price Index
Expected Relevance	Measures food affordability and local market pressure.
Known Limitations	Market coverage varies by country; basket index may introduce weighting bias.

Economic Conditions

Category	Description
Source Dataset(s)	World Bank GEM / IMF (proposed)
Raw Variables	Consumer Price Index (CPI) and Inflation Rate
Transformation Rule	Extract monthly national indicators, calculate monthly changes and assign them to all districts within each country.
Engineered Features	Monthly % change in CPI; Monthly Inflation Rate
Expected Relevance	Captures changes in purchasing power and the wider economy.
Known Limitations	The proposed variables are reported at the national level and therefore do not capture economic variation between administrative regions.

Humanitarian Access and Response

Category	Description
Source Dataset(s)	ReliefWeb API
Raw Variables	Publication date, report text, humanitarian themes, source and location
Transformation Rule	Aggregate reports by Admin 2 and month. Count reports by theme and derive NLP-based indicators (future work).
Engineered Features	Report count; Report count by classification
Expected Relevance	Captures detailed humanitarian activity and emerging crises not represented by structured datasets.
Known Limitations	Variable geographic precision and reporting bias.